



Soknarith (Soki) Sem

☎ (401) 952-6819 — ✉ seminaia401@gmail.com — 🌐 [Seminaia](#) — in [soknarith-sem](#) — 📍 Worcester, MA

SUMMARY

Passionate materials engineer with hands-on experience in material characterization, defect analysis, and process optimization within cGMP environments. Proficient in statistical methods, atomistic simulations, and scientific computing. Seeking to apply materials science to manufacturing support, material processing, and process improvements.

WORK EXPERIENCE

Research Assistant & Graduate Teaching Assistant May 2024 to Present
Worcester Polytechnic Institute - Worcester, MA

- Quantified defect energetics and transport behavior in oxide perovskites using DFT and thermodynamic modeling.
- Analyzed structure–property–process relationships across temperature and chemical potential regimes for manufacturing semiconductors.
- Computed temperature-dependent free energies and defect formation energies using Monte Carlo methods to predict equilibrium defect concentrations and transport properties.

Chemical Process Engineer May 2023 to May 2024
Knowles Precision Devices - New Bedford, MA

- Led R&D-to-production transition for high-voltage capacitor dielectric fluids.
- Optimized manufacturing layout and process flow using LEAN principles and statistical process controls, reducing production costs by 15%.
- Authored SOPs and work instructions for material handling and process scale-up, for training technicians and operators.
- Performed root cause analysis on material-related issues, including deviation investigation and CAPA implementation.

R & D Technician Mar 2023 to May 2023
Factorial Energy - Woburn, MA

- Fabricated Li-metal batteries in controlled glovebox and dry-room environments.
- Reduced material waste by 20% through inventory system redesign.

Downstream Purification Engineer Jul 2022 to Mar 2023
Eurofins - Framingham, MA

- Downstream purification of therapeutic proteins, with experience in chromatography and filtration techniques.
- Purified therapeutic proteins using AKTA Avant systems with Unicorn software achieving 95-99% purity.
- Purification of antibodies and enzymes using various chromatography techniques.
- Utilized DOEs to optimize buffer conditions and flow rate for maximum yield and purity.
- Extended TFF membrane lifetime by 30% through lifetime studies.

EDUCATION

M.S. Materials Science & Engineering GPA: 3.50
Worcester Polytechnic Institute - Worcester, MA 2024 to 2026

Relevant coursework: Thermodynamics of Materials, Advanced Process Engineering, X-Ray Diffraction

B.S. Chemical Engineering & B.S. Applied Mathematics GPA: 3.16
University of Rhode Island - Kingston, RI 2019 to 2022

Minors: Chemistry, Physics

TECHNICAL SKILLS

Atomistic Simulation: DFT (VASP & GPAW), Molecular Dynamics (LAMMPS)
Scientific Computing: Python, MATLAB, ASPEN Plus Dynamics, Linux, LaTeX, R, Unicorn
Laboratory: Glovebox, clean-room, HPLC, AKTA FPLC, TFF, SEM, XRD, DSC/TGA, UV-Vis, Osmometry, HPLC/MS, MS/MS, pH-meter, SDS-PAGE, KF titration meter
Manufacturing: SPC, DOE, Lean/Six Sigma, Process optimization, SOP development, Tech Transfer (MSAT), cGMP